FOURTH LESSON

Particles

Within the universe described in the previous lesson, light and things move. Light is made up of photons, the particles of light intuited by Einstein. The things we see are made of atoms. Every atom consists of a nucleus surrounded by electrons. Every nucleus consists of tightly packed protons and neutrons. Both protons and neutrons are made up of even smaller particles which the American physicist Murray Gell-Mann named ‘quarks’, inspired by a seemingly nonsensical word in a nonsensical phrase in James Joyce’s *Finnegans Wake*: ‘Three quarks for Muster Mark!’ Everything we touch is therefore made of electrons, and of these quarks.

The force that ‘glues’ quarks inside protons and neutrons is generated by particles that physicists, with little sense of the ridiculous, call ‘gluons’.

Electrons, quarks, photons and gluons are the components of everything that sways in the space around us. They are the ‘elementary particles’ studied in particle physics. To these particles a few others are added, such as the neutrinos which swarm throughout the universe but have little interaction with us, and the ‘Higgs bosons’ recently detected in Geneva in CERN’s Large Hadron Collider. But there are not many of these, fewer than ten types in fact. A handful of elementary ingredients that act like bricks in a gigantic Lego set, and with which the entire material reality surrounding us is constructed.

The nature of these particles, and the way they move, is described by quantum mechanics. These particles do not have a pebble-like reality but are rather the ‘quanta’ of corresponding fields, just as photons are the ‘quanta’ of the electromagnetic field. They are elementary excitations of a moving substratum similar to the field of Faraday and Maxwell. Minuscule moving wavelets. They disappear and reappear according to the strange laws of quantum mechanics, where everything that exists is never stable, and is nothing but a jump from one interaction to another.

Even if we observe a small empty region of space, in which there are no atoms, we still detect a minute swarming of these particles. There is no such thing as a real void, one that is completely empty. Just as the calmest sea